TECHX

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Meet Chintu

Automating Data Lineage on Azure for $0.04 a Pipeline

A multi-agent system built with LangGraph and Azure OpenAI — and the four walls we hit before it worked.

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IT'S ME

Sarveshwaran Rajagopal

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Enterprise AI & data engineering

Building AI and data platform systems, where anything that ships has to survive cost review, security approval and production.

Agentic systems in production

LangGraph, CrewAI, RAG pipelines and multi-agent orchestration — including Chintu, the system in this talk.

Practitioner, then teacher

Hands-on AI engineering content for a large developer community, built from systems that actually run.

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It Started With One Question in a Review

A senior leadership director, in a routine platform review

"Which source systems feed our churn model?"

Reasonable question. Senior person. A room full of people who had built the thing. And not one of us could answer it with confidence.

IT WAS DOCUMENTED

We had lineage documentation

Written carefully, by good engineers. Reviewed. Signed off.

IT HAD DRIFTED

Dozens of changes had landed since

The document described a system that no longer existed.

THE TRUTH WAS IN PEOPLE

Accurate lineage lived in two heads

And memory does not scale. Memory also goes on leave.

Data lineage: which raw table did this number come from, what happened to it on the way, and what breaks downstream if the source changes?

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Let Me Explain This the Way I Explain It at Home

For the next forty minutes, our data platform is a restaurant. Everything else follows from that.

IN THE KITCHEN



The restaurant



The signature dish



The ingredients



Ingredient traceability



The recipe book



Cost per plate

IN OUR PLATFORM



Our data platform



The churn model



Source tables



Data lineage



Our documentation



Cost per pipeline

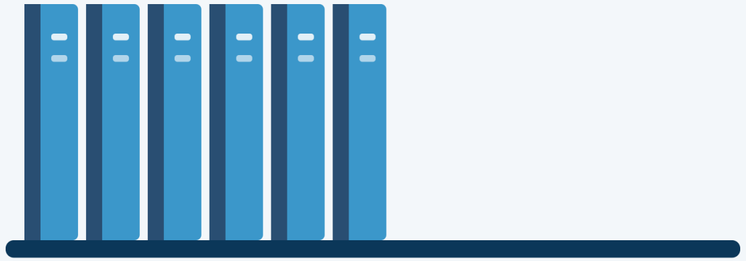
Remember cost per plate. It is the number that eventually saved this project.

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The Recipe Book Was Fiction

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WHAT THE DOCUMENTATION SAID



Six source tables

Written eighteen months ago. Never opened again.

WHAT WAS ACTUALLY RUNNING



Eleven source tables

Two created last quarter. One already deprecated.

Documentation is a snapshot. A data platform is a movie.

1,400	~3 hrs	4,200	Then
pipelines in scope	to trace one by hand	engineer-hours, once	start again — it changed

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ACT 2 — THE FIRST ATTEMPT

What My Team Tried First

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We are an AI team. So of course our first instinct was: throw it at a model.

THE TWO-HOUR PROTOTYPE

1. Read the entire SQL file
2. Paste all of it into the prompt
3. Ask for the lineage in plain English
4. Print whatever comes back

No chunking. No output schema. No routing. No cost tracking. Biggest model, every single time.

AND IT WORKED.

- ✓ Correct source tables, first attempt
- ✓ Readable transformation descriptions
- ✓ Caught a PII column we had missed
- ✓ Two hours, start to finish

The demo went down very well indeed.

This is where most AI talks end. This is where ours actually began.

Everything after this slide is the part nobody demos.

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Then Four Walls Appeared

WALL 1

The Bill

Our cost per plate came out higher than simply hiring people to do the tracing.

WALL 2

The Kitchen

We sent 1,400 orders in at once. The kitchen stopped taking orders at 300.

WALL 3

The Recipe

"Our production SQL is not going to a third-party model." End of conversation.

WALL 4

The People

Leadership wanted a number. The team wanted to know if they were being replaced.

Only one of these four walls was a technology problem.

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Cost Per Plate

WHAT WE PROJECTED, PER MONTH

~$8,200

1,400 pipelines, re-run whenever one changed, biggest model, long prose output

WHAT IT WAS REPLACING

~$5,600

The equivalent manual tracing effort at a loaded engineering rate

Our AI solution cost more than the problem it was solving.

WHY LEADERSHIP PAUSED IT IN WEEK THREE

Not because it failed. Because nobody in the room could explain the cost per plate — and a recurring bill that nobody can explain is a governance problem long before it is a finance problem.

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What Actually Cut the Bill

THE BIGGEST SINGLE WIN

Stop asking for prose. Ask for a schema.

Before: "Explain the lineage in detail." → three paragraphs back. We used one line of it.

After: "Return ONLY this JSON schema." → straight into the catalog, same shape every run.

3,100 → 900

output tokens per pipeline. Accuracy did not move.

2 **Cap the response length**

A runaway response is a runaway invoice. Hard ceiling on every call.

3 **Cache the fixed part**

Schema and instructions were identical on all 1,400 calls. Only the SQL changed.

4 **Cook it overnight**

Nobody needed pipeline 900 traced in two seconds. Trade speed for a discount.

5 **Send only what is needed**

Parse first. Send the joins and the select list, not 400 lines.

We were ordering full meals and eating only the rasam.

Not one of these five is an AI technique. This is ordinary engineering discipline, applied to a new kind of bill.

The Night the Kitchen Stopped Taking Orders

We had 1,400 pipelines, the work was parallel, and we were feeling confident.



429 · TOO MANY REQUESTS

Not a bug in our code. The kitchen's speed limiter.

AND THE RETRIES MADE IT WORSE

The job died at 300 of 1,400 — and we had paid for every one.

THE LIMIT THAT ACTUALLY BINDS

Tokens per minute — not requests per minute.

A queue, with a limit

Back off, politely

Remember where you stopped

Cook off-peak

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So We Stopped Prompting, and Built Chintu

Chintu is not a chatbot. He is a kitchen assistant we trained ourselves — and he only ever bothers the head chef when he genuinely does not know.



1 He preps in our kitchen

The scrubber strips values and credentials before anything leaves. The model never sees an ingredient — only the recipe structure.

2 He knows who should cook it

We had been sending the head chef to slice one onion. Chintu scores difficulty locally, then picks the cook.

3 He only asks when unsure

High-confidence traces go straight through. He interrupts a human only when he cannot tell. Roughly one plate in six.

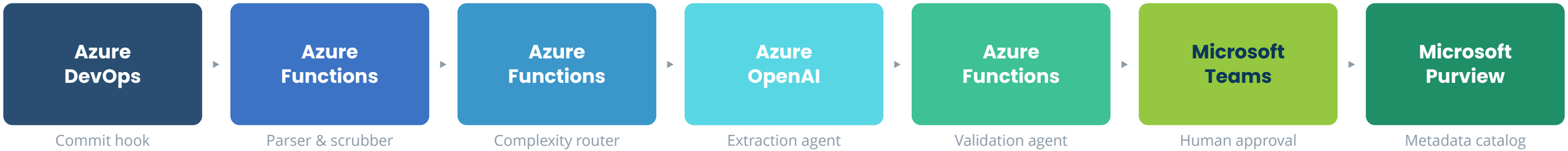
Chintu is not clever. Chintu is disciplined. That turned out to be the more useful quality.

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How Chintu Runs in Production on Azure

The same seven steps from the last slide — and where each one actually runs in our Azure tenant.



1 It runs inside our network

LangGraph runs on Azure Functions in our VNet. The scrubber runs first, and only structure reaches Azure OpenAI, over a private endpoint.

2 It can pause without forgetting

Graph state is checkpointed to Cosmos DB. A flagged trace waits for a Teams approval, then resumes exactly where it stopped.

3 Every plate is costed

Keys live in Key Vault. Tokens, tier and cost for every pipeline go to Application Insights. That is where four cents is measured.

Nothing exotic. Every box is a managed Azure service our platform team already runs.

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What Chintu Delivered

~$0.04

cost per plate
Down from ~$0.42

11 days

to clear the backlog
Against 4,200 manual hours

~1 in 6

plates need the chef
The rest go straight through

0

ingredients sent out
Structure only, by design

94%

of traces approved with no edit at all
Accuracy is a model metric. Approval rate is an adoption metric. Only one of them tells you the system is actually being used.

THE THING WE DID NOT EXPECT

Chintu found lineage that no document had ever captured — and one upstream table that had been feeding a live model long after everybody believed it was retired. Nobody asked him to look for that.

▶ **LIVE DEMO — Chintu, running. Three pipelines. You decide whether to approve the one he is unsure about.**

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What Actually Crosses the Pass

This is the diagram I drew on a whiteboard in the security review.

STAYS IN OUR KITCHEN

- ✓ Production databases
- ✓ Customer records and PII
- ✓ Literal values in queries
- ✓ Credentials and connection strings
- ✓ The parser and pre-processor
- ✓ The approved lineage catalog

Zero rows. Zero customer data.

GOES TO THE MODEL

- → Table and column names
- → Data types
- → Join and CTE structure
- → Transformation expressions

Structure only.

This one diagram did more for approval than every business case I wrote.

The Leadership Conversation

WHAT I SAID THE FIRST TIME

"We can use an LLM to automatically extract data lineage from our SQL pipelines."

WHAT I SAID THE SECOND TIME

"We can clear a 4,200-hour tracing backlog for about four cents a pipeline, with nothing leaving our network."

What they actually heard

- ✕ A recurring cost with no ceiling
- ✕ A new vendor dependency
- ✕ A compliance question they now own
- ✕ Another initiative that may not land

Paused in week three.

What changed

- ✓ A known cost per plate, not an open bill
- ✓ A named business outcome
- ✓ The data question answered upfront
- ✓ A measurable success criterion

Approved in one meeting. Then showcased twice.

Same project. Same technology. Different sentence.

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ACT 9 — WHAT I WOULD TELL YOU

What Changed in My Head

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FROM		TO
Start with the model	→	**Start with the painful task**
Ask what AI can do	→	**Ask what the task costs today**
Biggest model, always	→	**Cheapest cook who clears the bar**
Measure accuracy	→	**Measure approval rate and adoption**
Ask forgiveness on data	→	**Draw the boundary before you build**
Pitch the technology	→	**Pitch the cost per plate**

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Chintu is not clever. He is disciplined.

He knows what he costs. He knows what he is allowed to see. And he knows when to ask a human.

That was the whole trick.

The winners in the agentic era won't be the organisations with the most AI tools.

They'll be the ones who know what their AI costs, where their data goes, and who signs off when it gets it wrong.

Thank you · Questions?

Sarveshwaran Rajagopal · Senior Research Scientist, NIQ

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BEFORE YOU GO

Want to Build Systems Like Chintu?

Everything in this talk — the graph, the guardrail, the cost discipline — taught end to end.

FREE LIVE SESSION

Scan to join

No charge. Bring your laptop.

Build AI agents, step by step

From one tool call to a multi-agent graph — and the failure modes in between.

Work on real business use cases

The shape of problem Chintu solved, not a toy notebook.

Move into AI with practical skills

What to build, what to measure, and what survives a review.

FIND ME HERE

Let's Stay in Touch

Slides, the demo code and the full write-up — happy to share.

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